

- 6 A rectangular coil PQRS of dimensions 14.0 cm by 12.0 cm, moves with a constant speed of 2.0 cm s^{-1} through a region of uniform magnetic field WXYZ of width 10.0 cm, as shown in Fig. 6.1. There is a magnetic flux density B of 1.5 T in WXYZ directed into the plane of the paper.

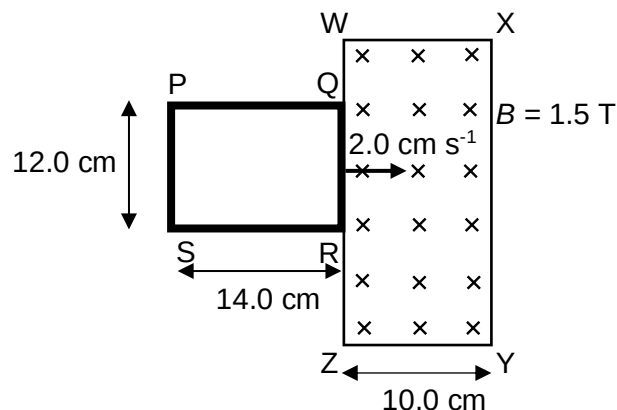


Fig. 6.1

- (a) State and explain the direction of induced current in the coil PQRS when the coil enters the field. [3]

Method 1: As the coil enters the field, there is an increase in magnetic flux linkage into the plane of the coil. By Faraday's Law, an e.m.f. [1] and a current are induced in the coil. By Lenz's Law, the induced current flows in an anti-clockwise direction [1] so as to create a magnetic field pointing out of the plane of the coil to oppose the increase in flux linkage. [1]

Method 2: As the coil moves to the right, electrons along QR moves to the right. This means that a current moves to the left [1], and using Fleming's Left Hand Rule, the electrons in QR experience a downward magnetic force [1], causing the electrons to travel in a clockwise direction in the coil. This means that an induced current is travelling in an anti-clockwise direction in the coil. [1]

- (b) Calculate the maximum possible magnetic flux linkage during the motion.

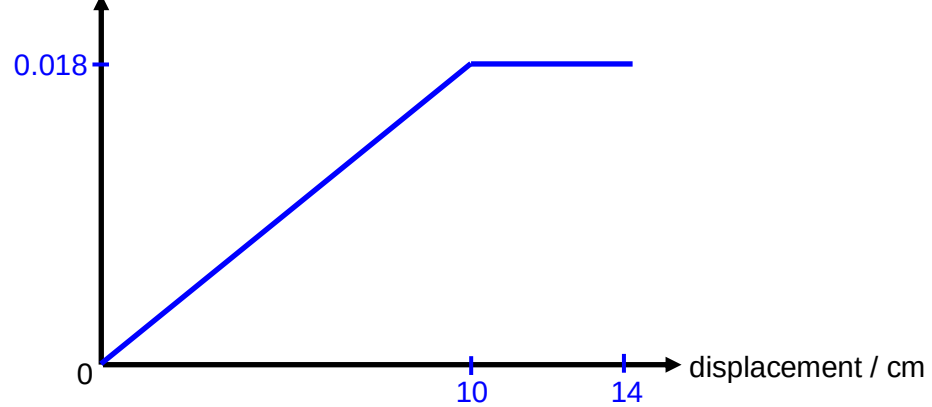
$$\phi_{\text{max}} = BA = 1.5 \times 0.120 \times 0.100 = 0.018 \text{ Wb}$$

maximum flux linkage = Wb [1]

(c) The coil PQRS is moved from the position shown in Fig. 6.1 until side PS is aligned with WZ. Sketch and label with appropriate values on the axes, the following graphs:

(i) the variation with displacement of magnetic flux linkage through the coil. [2]

magnetic flux linkage / Wb



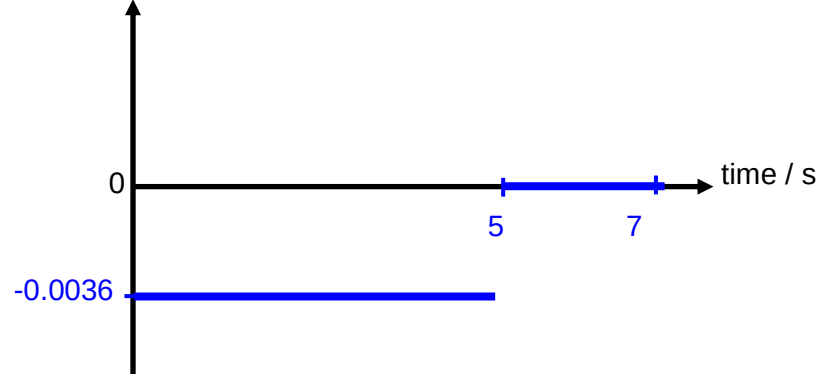
Straight line starts from origin to (10, 1.8) [1]

Horizontal line from (10, 1.8) to (14, 1.8) [1]

Minus one mark if no/wrong label

(ii) the variation with time of induced e.m.f. in the coil. [2]

e.m.f. / V



Horizontal line with e.m.f. = -0.0036 V, starting from 0 to 5 s [1]

Horizontal line with e.m.f. = 0 V, starting from 5 to 7 s [1]

Minus one mark if no/wrong label

7 A light elastic band of length 20 cm is stretched to a length of 28 cm and then allowed to